TEXORA · LECTURE SHEET

Destructive and Non-Destructive Testing

Sacrificing samples to learn the truth, and interrogating parts without harming them: the destructive test family, the six great NDT methods — visual, penetrant, magnetic particle, ultrasonic, radiography, eddy current — and how to choose between them.

ME-1207 · ENGINEERING MATERIALS · TOPIC 5 · LEVEL 1 TERM 2

1. Two philosophies

Destructive testing (DT) loads a specimen to failure and measures real numbers — strength, ductility, toughness (Topic 2's whole toolkit: tensile, impact, fatigue, creep, bend and torsion tests). The specimen dies, so DT answers for a *batch* statistically, never for the exact part that will fly, hold pressure or carry a load. **Non-destructive testing (NDT)** hunts defects — cracks, voids, inclusions, thin walls — in the actual part, which then goes into service. Industry uses both in harness: DT qualifies the material and process, NDT verifies every delivered piece. Welds in pressure vessels, boiler tubes in a dyehouse, shafts and gears in textile machinery all live under this regime.

2. The NDT family

METHOD	PRINCIPLE	FINDS	LIMITS
Visual (VT)	Eyes, borescopes, gauges, good light	Surface defects, misalignment, corrosion — always the first method	Only what can be seen
Dye penetrant (PT)	Low-viscosity dye seeps into surface-breaking cracks by capillary action; developer draws it out as a coloured bleed	Surface cracks in *any* non-porous material	Surface-breaking flaws only; useless on rough/porous surfaces
Magnetic particle (MT)	Magnetise a ferromagnetic part; cracks distort the flux,	Surface and near-surface cracks in steels	Ferromagnetic materials only; needs demagnetising after

	and iron powder gathers at the leakage field		
Ultrasonic (UT)	High-frequency sound (0.5–25 MHz) pulse-echoes off internal boundaries; time of flight locates and depths the reflector	Internal flaws, laminations, weld defects, wall thickness (corrosion surveys without shutdown)	Needs couplant and skilled interpretation; coarse-grained castings scatter the beam
Radiography (RT)	X-rays or γ-rays shadow the part onto film/detector; voids absorb less and print darker	Volumetric flaws — porosity, slag, cracks in welds; gives a permanent image	Radiation safety regime; two-side access; tight cracks parallel to the beam can hide
Eddy current (ET)	An AC coil induces eddy currents in a conductor; flaws disturb them and the coil's impedance reports it	Surface/near-surface cracks, thickness, coating, alloy sorting; ideal for tubes	Conductive materials only; shallow penetration

Two supporting acts: **hydrostatic/pneumatic proof testing** — pressure the vessel above working level with water and watch for leaks or weeping (destructive only if it fails, which is the point), and **acoustic emission** — listening with sensors for the tiny clicks of growing cracks in a structure under load.

3. Choosing a method

Match the method to the flaw's address. Surface-breaking and non-magnetic → penetrant (aluminium, stainless, ceramics, plastics). Surface, ferromagnetic → magnetic particle, faster and more sensitive. Buried inside a weld → ultrasonic or radiography: UT for thick sections, exact depth and no radiation hazard; RT when a permanent film record is required or geometry favours it. Tubes and bars on a production line → eddy current, automated at speed. And remember what NDT cannot do: it reports defects against acceptance criteria, but it does not measure tensile strength — the number still comes from destroying samples (Section 1), however good the pictures get.

4. Key points

- DT measures true properties on sacrificed samples; NDT finds flaws in the real part. Both are needed.
- PT = any material, surface-breaking only. MT = ferromagnetic only, surface/near-surface. ET = conductors only, shallow.

- UT and RT see inside: UT gives depth from echo time; RT gives a picture but carries radiation controls.
- Proof pressure testing and acoustic emission watch the whole structure at once.
- Choose by flaw location, material and required record — not by habit.

5. Practice

Answer in your own words; brief model answers follow.

1. A welded pressure-vessel seam may contain internal slag inclusions. Recommend two suitable NDT methods and one advantage of each.
2. Fine surface cracks are suspected on (a) a steel crankshaft and (b) an aluminium bracket. Choose a method for each and justify.
3. A corroded steam line must have its remaining wall thickness mapped while the plant keeps running. Which method, and how does it report thickness?
4. Why can a factory not replace tensile testing entirely with ultrasonic inspection, however carefully done?
5. A dye penetrant test on a rough sand-cast surface gives a confusing mass of indications. Explain the failure and name a better-suited method for that casting’s surface cracks if it is cast iron.
6. State what a hydrostatic test proves that a radiograph of the same vessel cannot.

Answer key

1. Ultrasonic testing — finds and depths internal reflectors in thick welds with no radiation hazard and only one-side access. Radiography — produces a permanent image of volumetric flaws like slag and porosity that inspectors and clients can review later. (Either alone is acceptable practice; together they cross-check.)
2. (a) Magnetic particle testing: steel is ferromagnetic, MT is fast, cheap and very sensitive to surface cracks — the powder gathers right on the flaw. (b) Dye penetrant: aluminium is non-ferromagnetic so MT is physically impossible; PT works on any non-porous metal and reveals the cracks as coloured bleed-outs.
3. Ultrasonic thickness gauging. A pulse is sent through the wall and the echo time from the far surface is measured; with the known sound velocity in steel the instrument converts time to thickness. It works from the outside, through a couplant dab, while the line stays in service.
4. Because UT measures echoes from boundaries and defects — geometry and sound behaviour — not the stress at which the material yields or breaks. Two batches can show identical ultrasonic responses yet differ in strength (different heat treatment, chemistry drift). Strength is a mechanical fact; it must be measured mechanically, on samples, at qualification and at intervals.

5. Penetrant bleeds out of every pore and scratch of a rough porous casting, flooding the developer with background until real cracks drown in noise — PT demands smooth, non-porous surfaces. For cast iron (ferromagnetic), magnetic particle testing is the better tool: it ignores mere roughness far better and picks out the crack leakage fields.
6. Actual leak-tightness and structural integrity under pressure, of the whole vessel as built — including its nozzles, gaskets and workmanship — demonstrated in one act. A radiograph shows the condition of the welds it imaged but says nothing about leaks at joints or behaviour under load; the hydro test is the vessel’s own proof, which is why codes require it after fabrication and major repair.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib